HAPPIEST HEALTH · DIGITAL TWIN CHALLENGE 2026

GlucoTwin

A reliability-aware digital twin for 120-minute glucose forecasting

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Code: github.com/Raju-09/GlucoTwin · Video: youtu.be/Vsmb3QibpKI (unlisted) · MIT license

Research prototype on synthetic data. Not a medical device.

THE PROBLEM

A reading every 5 minutes, but alerts that look only a short way ahead

101M

people with diabetes in India, per the ICMR-INDIAB study (*The Lancet Diabetes & Endocrinology*, 2023).

~20 min

typical look-ahead of commercial predictive alerts, such as Dexcom Urgent Low Soon. Many alerts are plain thresholds.

120 min

the horizon GlucoTwin explores: enough time for a meal change or a walk, if the forecast can be trusted.

Research question. Does a time-series model beat a persistence baseline on a leakage-safe held-out test, and when should the system withhold a forecast?

APPROACH

The virtual patient is three things at every time step

$$S(t) = (\theta p , ht , \delta t)$$

Static phenotype

EHR context · 9 features

Age, sex, BMI, baseline HbA1c, diabetes duration, historical fasting glucose, metformin, SGLT2 inhibitor and dawn-phenomenon flags.

Dynamic history

Causal 120-minute window

24 CGM lags at 5-minute cadence, rolling mean and spread, velocity and acceleration, circadian terms, plus simulated wearable signals.

Twin integrity

Stream quality state

Available, degraded or withheld, with a reason code, data freshness and a quality score. This decides whether a forecast is shown.

ARCHITECTURE

From raw streams to a gated forecast

1 · Ingest
CGM every 5 minutes, wearables, EHR (9 fields)

→

2 · Validate
Freshness, range, gap and jump checks

→

3 · Features
Lags, rolling stats, circadian: 41 / 50 / 56 by configuration

→

4 · Forecast
HistGradientBoosting, +120 min

→

5 · Gate and serve
Available, degraded or withheld; FastAPI; Streamlit

Simulated sensor layer

Interstitial lag (τ = 10 min), AR(1) noise and a rate clamp of 3.8 mg/dL/min mimic real CGM behavior.

Leakage control

Chronological split with a 120-minute buffer between training and test windows.

Cohort

60 synthetic type 2 diabetes patients, 224,681 windows, fixed generator seed.

RELIABILITY GATE

Before any forecast is shown, the input stream must pass a check

State	Conditions	Behavior
Available	Data fresh (under 15 min), full 24-step history, values within 40–400 mg/dL	Forecast shown
Degraded	Latency of 15–20 min or minor packet loss; history intact	Forecast shown with a warning and reason code
Withheld	Stale stream (over 20 min), gap over 30 min, sensor dropout, or a jump above 4.0 mg/dL/min	No number returned, only an error code

Fail closed. The gate is rule-based and checks the data, not the model. The risk–coverage trade-off, meaning how much error is avoided for how many forecasts withheld, is evaluated in scripts/evaluate_risk_coverage.py.

RESULT 1 · ABLATION

EHR context helps within patients; wearables add nothing here

Configuration	MAE	Δ vs CGM only	95% CI of Δ
Persistence baseline	46.88	+28.57	n/a
Linear extrapolation (30-min slope)	86.83	+68.52	n/a
A · CGM only (41 features)	18.31	reference	n/a
B · CGM + EHR (50)	17.61	−0.70	[−0.96, −0.49]
C · CGM + wearables (47)	18.32	+0.01	[−0.08, +0.07]
D · All streams (56)	17.67	−0.64	[−0.88, −0.44]

MAE in mg/dL at +120 min. Chronological hold-out, days 12–14, 43,888 windows, 60 patients. Training and test share the same patients, so this is a within-patient test. Wearable features gave no measurable gain under this simulator; we report that as a finding.

RESULT 2 · TESTING OUR OWN CLAIM

On unseen patients, EHR context gave no benefit

CGM only

17.65

mg/dL, mean LOPO MAE

CGM + real EHR

17.65

Δ +0.01, p = 0.956

CGM + shuffled EHR

20.47

Δ +2.82 mg/dL, p < 0.001

Leave-one-patient-out, 60 patients, 224,681 windows. The within-patient gain did not transfer to new patients.

Shuffling EHR across patients made predictions worse, which is consistent with the model using EHR as a patient-specific signal. We have not yet tested why; relating the error penalty to the distance between true and swapped EHR is the planned next check.

RESULT 3 · THRESHOLD EVENTS AT +120 MIN

High glucose is detected far better than the baseline; lows are missed

Event	Model	Positive windows	Sensitivity	Precision	False alarms / day
Above 180 mg/dL	Persistence	12,246	26.6%	25.8%	61.4
Above 180 mg/dL	GlucoTwin	12,246	76.2%	75.8%	19.6
Below 70 mg/dL	Persistence	71	12.7%	12.7%	0.4
Below 70 mg/dL	GlucoTwin	71	0.0%	0.0%	0.0

High glucose. Sensitivity rose from 26.6% to 76.2% and false-alarm windows per day fell from 61.4 to 19.6. About 28% of test windows are above 180, so this is a common state, not a rare spike.

Low glucose. The model detected none of the 71 windows below 70. A mean-seeking regression model rarely predicts extremes, so this needs a different loss.

Counts are 5-minute windows, not distinct episodes; the test set spans 152.4 simulated patient-days. Synthetic held-out test; not evidence of clinical effectiveness.

CLINICIAN DASHBOARD

A doctor can inspect the virtual patient and break the data stream

Patient state view

60 synthetic patients, a 14-day timeline slider, the three-part state S(t), and badges that show where each value came from.

Failure injection

Inject a stale reading, a sensor gap, a 600 mg/dL spike or packet dropouts and watch the gate move to degraded or withheld.

Serving and tests

FastAPI latency over 100 calls: median 7.8 ms, p95 11.1 ms, p99 13.3 ms. 69 unit tests pass. Docker Compose setup included.

Live demo: streamlit run frontend/streamlit_app.py

LIMITS AND NEXT STEPS

What this does not show yet, and how we would find out

Known limits

- All data is synthetic; no real-patient performance is claimed.
- No lows detected: 0 of 71 windows below 70 mg/dL.
- The EHR benefit did not transfer to unseen patients.
- Wearables added no measurable value in this simulator.
- Forecasts are point estimates without uncertainty intervals.

Next steps

- Validate on open CGM datasets under their access terms.
- Add a time-of-day baseline and a new-onset event analysis.
- Report alarms as episodes with a refractory period.
- Add calibrated prediction intervals tied to the gate.
- Train with an asymmetric loss for low glucose.

Forecast earlier. Intervene smarter. Withhold when the data cannot be trusted.